SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

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QUESTIONS: 05 **Total Marks:50**

Annexure A

Pseudocode Writing Task (Algorithm Design)

Criteria	Problem Understanding *(2)*	Algorithm Design *(3)*	Use of Control Structures *(2)*	Pseudocode Structure *(1)*	Completeness *(2)*	***Total(10)***
Weightage	*20%*	30%	20%	10%	20%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Annexure A

Q1. Dataset: Online Shopping Transactions

Order ID	Products Purchased
O1	Mobile, Earphones
O2	Mobile, Charger
O3	Earphones, Charger
O4	Mobile, Earphones, Charger
O5	Mobile, Earphones

Question:
Design pseudocode for the **Apriori algorithm** to discover frequent product combinations from the online shopping dataset using a specified minimum support threshold. Clearly illustrate the candidate itemset generation and pruning process at each iteration.

Q2. Dataset: Library Book Borrowing Records

Borrow ID	Books Borrowed
B1	Data Science, Python
B2	Python, Machine Learning
B3	Data Science, Machine Learning
B4	Data Science, Python, Machine Learning
B5	Python, Machine Learning

Question:
Develop pseudocode for the **FP-Growth algorithm** to identify frequent borrowing patterns without generating candidate itemsets. Explain the steps involved in constructing the FP-Tree using the given borrowing records. *(BL4)*

Q3. Dataset: Frequently Purchased Services

Service Combination	Support Count
{Internet}	4
{Streaming}	4
{Cloud Storage}	3
{Internet, Streaming}	3

{Streaming, Cloud Storage} 2

Question:
Write pseudocode to generate **association rules** from the given frequent service combinations. Include the procedures for confidence calculation, rule generation, and filtering based on a minimum confidence threshold. *(BL3)*

Q4. Dataset: Student Course Enrollment

Student ID	Courses Enrolled
S1	Python, Database
S2	Python, AI
S3	Database, AI
S4	Python, Database, AI
S5	Python

Question:
Design pseudocode for **constraint-based frequent itemset mining** where only itemsets containing the course **"Python"**are considered during the mining process. Explain how the constraint affects candidate generation and pruning. *(BL4)*

Q5. Dataset: Hospital Treatment Hierarchy

Patient ID	Treatments Received
P1	Healthcare, Cardiology, ECG
P2	Healthcare, Orthopedics
P3	Healthcare, Cardiology
P4	Healthcare, Cardiology, ECG
P5	Healthcare, Orthopedics

Question:
Develop pseudocode for **multilevel association rule mining** using the treatment hierarchy provided in the dataset. Explain how rules are generated at different hierarchy levels and how concept hierarchies improve pattern discovery.

Q1. ANSWER:

Pseudocode:

```
BEGIN
  Read transactions and minimum support minsup
  C1 <- all 1-itemsets
  k <- 1
  WHILE Ck is not empty DO
      Count support of every candidate in Ck
      Lk <- candidates whose support >= minsup
      Generate C(k+1) by joining itemsets in Lk
      Prune candidates having any infrequent subset
      k <- k + 1
  END WHILE
  RETURN all frequent itemsets L1, L2, ...
END
```

For the given data, C1 contains Mobile, Earphones and Charger. Candidate pairs are generated from these items. With each iteration, candidates below the minimum support are removed before the next candidate set is formed. This join-and-prune process reduces unnecessary support counting.

Q2. ANSWER:

Pseudocode:

```
BEGIN
  Read transactions and minimum support minsup
  Count frequency of each item
  Remove items with support < minsup
  Sort remaining items in each transaction by descending frequency
  Create an empty FP-Tree and header table
  FOR each transaction DO
      Insert its ordered items into the FP-Tree
      Increment counts on existing nodes
      Create new nodes when required
  END FOR
  Mine the FP-Tree recursively using conditional pattern bases
  Build conditional FP-Trees and extract frequent patterns
  RETURN frequent itemsets
END
```

For the records, Python and Machine Learning occur frequently, while Data Science occurs in three transactions. The FP-Tree shares common prefixes, so repeated paths are stored compactly. Unlike Apriori, FP-Growth does not repeatedly generate candidate itemsets.

Q3. ANSWER:

Pseudocode:

```
BEGIN
  Read frequent itemsets F and minimum confidence minconf
  FOR each frequent itemset I in F DO
      Generate all non-empty proper subsets A of I
      FOR each subset A DO
          B <- I - A
          confidence <- support(I) / support(A)
          IF confidence >= minconf THEN
              Output rule A -> B with its support and confidence
          END IF
      END FOR
  END FOR
END
```

For {Internet, Streaming} with support count 3, possible rules are Internet -> Streaming and Streaming -> Internet. Their confidence is calculated using the support count of the antecedent. For {Streaming, Cloud Storage}, the same procedure is applied. Only rules meeting the specified minimum confidence are retained.

Q4. ANSWER:

Pseudocode:

```
BEGIN
  Read transactions and minimum support minsup
  C1 <- all items
  Remove every item except Python
  L1 <- frequent Python itemset if support(Python) >= minsup
  Generate only candidate itemsets that contain Python
  Count their support
  Prune candidates whose support < minsup
  Repeat join-and-prune until no more candidates exist
  RETURN frequent itemsets containing Python
END
```

For the dataset, Python occurs in S1, S2, S4 and S5, so support(Python) = 4/5 = 80%. Candidate itemsets such as {Python, Database} and {Python, AI} are considered, while itemsets without Python are never generated. Thus the constraint reduces candidate generation, support counting and memory usage.

Q5. ANSWER:

Pseudocode:

```
BEGIN
  Define hierarchy:
      Healthcare
        -> Cardiology -> ECG
        -> Orthopedics
  For each hierarchy level DO
      Transform transactions to the concepts at that level
      Generate candidate itemsets
      Count support
      Prune itemsets below minimum support
      Generate association rules from frequent itemsets
  END FOR
  Return rules from general to specific levels
END
```

At the higher level, Healthcare is common to all 5 patients. At the next level, Cardiology occurs in P1, P3 and P4, while Orthopedics occurs in P2 and P5. At the more specific level, ECG occurs in P1 and P4. Thus rules can be discovered from broad treatment categories to specific treatments. Concept hierarchies improve pattern discovery by revealing both general and detailed relationships.